Hidden Markov Models

Garrett Lee, Isaac Herman, Trey Elder, Injun Lyo

STAT 5440 · Final Project

SECTION 01

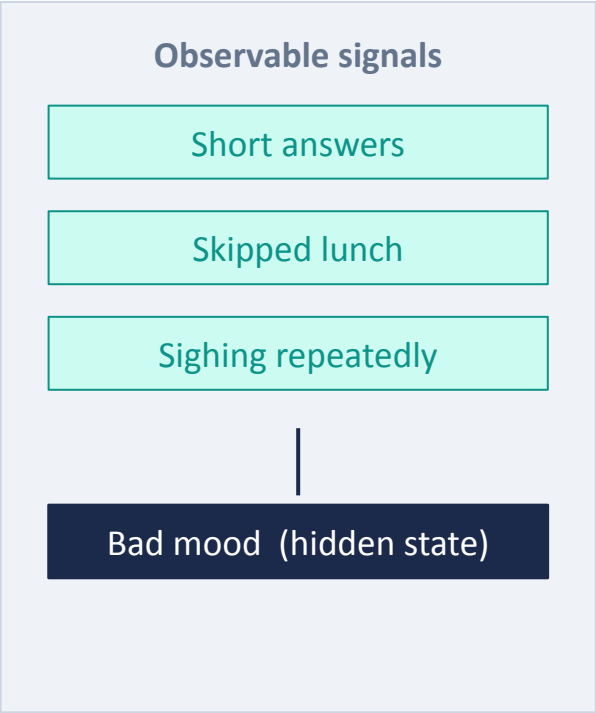
Motivation

Inferring latent performance states in MLB pitching

MOTIVATION

Inferring what we can't see

- You can't see your friend's mood directly
- From observable signals, we infer hidden states
- This is the core idea behind a Hidden Markov Model



MOTIVATION

What is a Markov model?

The next state depends **only on the current state** — not on any prior history

Transition matrix V

	Run (t+1)	Movie (t+1)
Run (t)	40%	60%
Movie (t)	70%	30%

Rows = current state · Columns = next state

If Bob ran today:

- 40% runs again tomorrow
- 60% watches a movie

If Bob watched a movie:

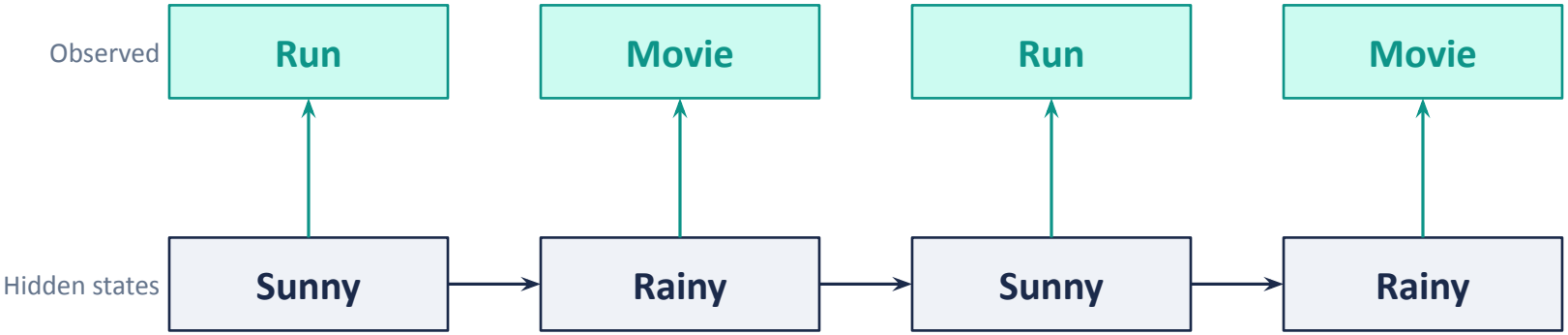
- 70% runs tomorrow
- 30% watches again

Bob's daily activity — tomorrow depends only on today

MOTIVATION

What is a Hidden Markov model?

- Observations are driven by an unobserved hidden state — not by previous observations
- The hidden state evolves over time as a Markov process



We only observe the emissions (top row) — the hidden states (bottom row) must be inferred

MOTIVATION

The three components of an HMM

01

Hidden States

Discrete, unobserved conditions that evolve over time via a Markov process

e.g. sunny / rainy

02

Transition Matrix

Probabilities of moving from one hidden state to another between time steps

$P(\text{rainy} \longrightarrow \text{sunny}) = 0.4$

03

Emission Probabilities

Probability of each observable outcome given the current hidden state

$P(\text{run} \mid \text{sunny}) = 0.6$

We only observe the emissions — the goal is to infer the hidden states and parameters that generated them

MOTIVATION

Why not standard regression?

1 The cause is invisible

Regression requires observed predictors — the hidden state driving outcomes has no column in the data

2 Independence assumption is violated

Sequential data is autocorrelated through the hidden state — regression assumes independence given covariates

3 One fixed relationship, but the process shifts

Each hidden state has its own emission distribution — regression can't fit separate models without knowing which state each observation belongs to

MOTIVATION

Real-world applications

SPEECH RECOGNITION

Hidden states: phonemes
Observations: acoustic signal

Backbone of the field before deep learning

GENOMICS

Coding vs. non-coding DNA regions inferred from nucleotide sequence patterns

FINANCE

Market regimes: bull, bear, high-volatility periods

Inferred from price and return sequences

SPORTS ANALYTICS

Latent performance states inferred from sequences of in-game actions

Our application today

SECTION 02

Model Specification

Inferring latent performance states in MLB pitching

THE MODEL

Likelihood

1 Independent Hidden Markov Sequences

Each game is an independent sequence of pitch-level four-seam fastball velocities. For each pitch t in game g, an unobserved state $Z_{g,t}$ comes from {Baseline, Hot}. Conditional on that state, velocity is modeled as a Normal draw with a state-specific mean and shared standard deviation.

2 First-Order Markov State Evolution

The hidden state evolves pitch-to-pitch as a first-order Markov process. The current state depends only on the immediately previous pitch's state, not the full history. Games are independent sequences: at the start of each game the state is drawn from an initial-state distribution.

3 Why Velocity? Why One Pitch Type?

We use velocity alone because it is the clearest interpretable signal of pitcher condition. Restricting to one pitch type prevents the hidden states from simply rediscovering pitch type instead of latent performance mode.

THE MODEL

Priors

1 Emission Means

Emission means for the two states get Normal priors centered at realistic values from Cole's pre-2024 fastball history. The lower-velocity state is centered near 96 mph, the higher near 99 mph, with a wide 2 mph standard deviation. This stabilizes estimation without overpowering the current season's data.

2 Transition Probabilities

Transition probabilities get Dirichlet priors that favor staying in the same state. This reflects the belief that pitcher condition is persistent pitch-to-pitch rather than flipping randomly. Dirichlet is the natural multi-category generalization of the Beta prior.

3 Initial State Distribution

The initial state distribution gets a Dirichlet prior that favors beginning a game in the Baseline state, matching the expectation that a pitcher is more likely to open in a typical rather than elevated performance state.

THE MODEL

Hierarchical Structure

1 Top Level: Global Parameters

State-specific velocity means, the transition matrix, and the initial state distribution. These are shared across all games and govern how states behave throughout the entire season.

2 Middle Level: Hidden States

Pitch-by-pitch hidden states that evolve within each game according to the transition matrix and reset at the start of each new game. States determine which velocity regime each pitch belongs to.

3 Bottom Level: Observed Velocities

Observed velocities generated from a Normal distribution centered at the mean corresponding to the current hidden state. The generative chain: Global parameters → States → Observed velocity. Note: this is a latent variable hierarchy, not a full multilevel random-effects model.

THE MODEL

Key Modeling Assumptions

01

Two Time-Homogenous Latent States

Exactly 2 meaningful latent performance states that differ only through average fastball velocity. Within-state noise level is constant across both states and all pitches.

02

Persistence

State changes depend only on the previous pitch, giving the model temporal persistence. Games are treated as independent sequences justified by several days of rest between outings, so no carry-over of latent state across starts.

03

Single Normal Emission

The emission distribution is assumed Normal and symmetric around a single mean per state. Fixing the variance and using one observable keeps the model stable, but means states capture only one dimension of pitcher condition.

04

Fixed Variance

Fixing the variance over latent states simplifies the Gibbs sampler and preserves conjugacy for the emission means. This also avoids introducing additional variance parameters

Labels are ordered so lower mean = Baseline, higher mean = Hot, ensuring interpretable posteriors across all iterations and chains

SECTION 03

Applied Example

Inferring latent performance states in MLB pitching

APPLIED EXAMPLE

Context

Two latent states: Baseline / Hot

A pitcher's true condition is unobserved. We only see the noisy velocity measurements that each state produces.

Single emission: release_speed (mph)

Restricting to one pitch type keeps the hidden state interpretable. Otherwise, the HMM discovers pitch type, not performance condition.

Games are independent sequences

Roughly 5 days between starts. The state resets at the beginning of each game; no carry-over modeled.

One pitcher, one partial season

Gerrit Cole, 2024: elbow injury kept him out the first half; he debuted June 19 and made 17 starts down the stretch. A comeback season means his velocity profile might oscillate more than usual, giving the HMM's multi-state structure more to work with. It might also test whether priors build on pre-injury data remain weakly informative. If the data can speak for itself, we should see the posterior shift away from prior beliefs.

AT A GLANCE

Gerrit Cole

RHP, New York Yankees

2024

17 starts (post-injury)

697

four-seam fastballs modeled (of 1,531 pitches total)

APPLIED EXAMPLE

Dataset

01

Statcast (the source)

MLB's official pitch-tracking system records release_speed, spin, movement, and pitch type for every pitch. Raw data served publicly through Baseball Savant.

02

baseballr (how we access it)

An R client for Baseball Savant. We call statcast_search() in weekly chunks across 2019–2024 so individual API calls stay small and are easy to retry.

RAW STATCAST PULL

All pitch types, all fields

Date	Pitch	Release Speed	Spin
2024-06-25	FF	96.4	2410
2024-06-25	SL	88.1	2690
2024-06-25	CH	87.9	1580
2024-06-25	FF	97.1	2425
2024-06-25	CU	83.0	2850

pitch_type · release_speed · spin · movement · extension · count …

RESHAPED FOR HMM

FF only, chronological, one metric

Season	Game	Pitch #	Release Speed
2024	1	1	95.8
2024	1	2	96.2
2024	1	3	96.9
2024	1	4	97.4
2024	1	5	96.1

Y = list of G games; Y[[g]] = ordered vector of FF velocities

Sort within game by inning, then at-bat, then pitch number to recover true chronological order.

APPLIED EXAMPLE

Three prior families

Every unknown in the HMM gets its own prior. Each encodes a belief about how pitchers perform.

01

EMISSION MEANS

$\mu_{\square}$ ~ Normal

Center each state's mean velocity at a reasonable starting point, with enough spread to let the data dominate.

02

TRANSITION ROWS

$v_{\square}$ ~ Dirichlet

Encode the belief that states are sticky. Most pitchers stay in the same state as the one before.

03

INITIAL STATE

π ~ Dirichlet

Assume pitchers typically open a start in Baseline rather than already elevated.

APPLIED EXAMPLE

Prior on the emission means

How we formed our priors

Used Cole's 2019–2023 four-seam velocities (pre-2024) as a reference, split at the 75th percentile:

- Bottom 75% → approximates the Baseline state
- Top 25% → approximates the Hot state

The 75/25 split is a heuristic, not a structural claim.

HISTORIC REFERENCE (2019-2023 FF)

State	Mean (mph)	SD (mph)	N (pitches)
Baseline	96.71	1.03	5,388
Hot	98.85	0.56	1,756

PRIOR SPECIFICATION

State	Prior mean	Prior SD
Baseline	96 mph	2 mph
Hot	99 mph	2 mph

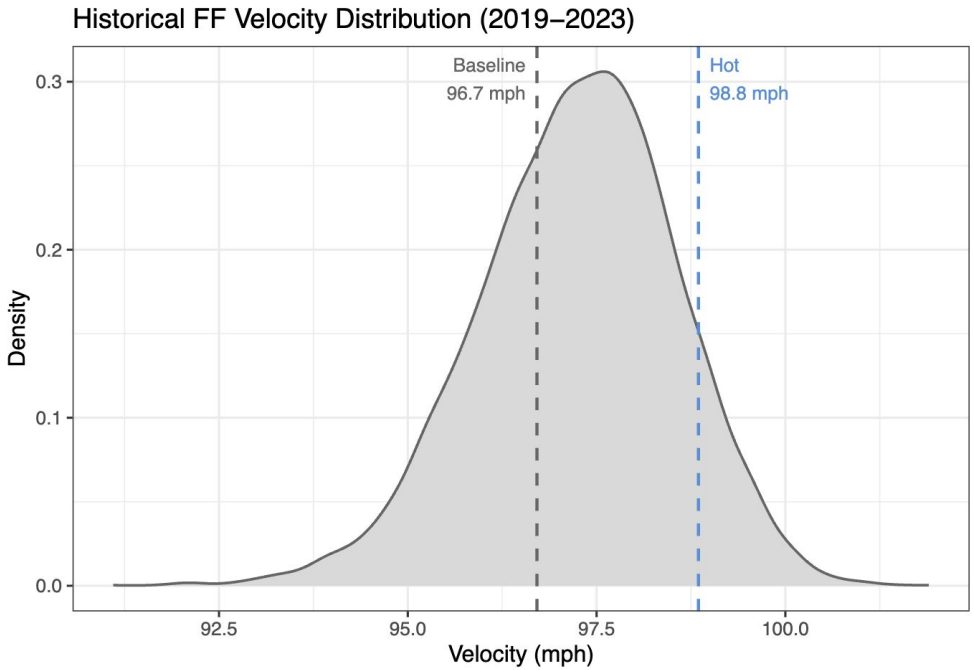
$$\mu_Baseline \sim N(96, 2^2)$$
$$\mu_Hot \sim N(99, 2^2)$$

σ is fixed at the empirical SD of Cole's 2024 four-seam velocities (not sampled)

APPLIED EXAMPLE

Historic FF velocities (2019–2023)

Distribution of Cole's pre-2024 four-seam velocities, split at the 75th percentile into Baseline (bottom 75%) and Hot (top 25%). The approximate means of these two groups become the centers of the emission-mean priors.



APPLIED EXAMPLE

Prior on the transition matrix

Dirichlet concentration to stay probability

Each row of v gets a Dirichlet prior. Its parameter vector tells the sampler how much prior weight to place on each outcome. Larger numbers mean that outcome is more likely a-priori.

For Baseline's row we set Dir(8, 1):

P(stay) = 8 / (8 + 1) ≈ 89%

The 8-to-1 ratio encodes the prior belief that states are sticky. Most pitches stay in the same state as the one before.

Why sticky?

Without a sticky prior, the sampler can flip states every noisy pitch, turning the hidden sequence into noise rather than a regime-switching signal.

TRANSITION PRIOR v

	→ Baseline	→ Hot
Baseline	**~89%**	~11%
Hot	~11%	**~89%**

Rows = current state · Columns = next state

Dir(8, 1) on each row
~89% stay, ~11% switch

APPLIED EXAMPLE

Prior on the initial state

Where does a start begin?

Before each game begins, we need a belief over the pitcher's opening state. We set $\pi \sim \mathrm{Dir}(6, 2)$.

From concentration to probability

$P(\text{start} = \text{Baseline}) = 6 / (6 + 2) = 75\%$
$P(\text{start} = \text{Hot}) = 2 / (6 + 2) = 25\%$

Rationale

The asymmetry reflects common sense: pitchers typically open in ordinary form rather than at peak velocity. The Dir(6, 2) concentration is weakly informative, so any given start's opening-state posterior can still shift toward Hot if the data support it.

Note on game independence

Because games reset, π governs every one of the 17 starts (not just the season opener).

INITIAL DISTRIBUTION π

$$\pi \sim \mathrm{Dirichlet}(6, 2)$$

APPLIED EXAMPLE

Gibbs sampler

The hidden states and the parameters depend on each other. The Gibbs sampler breaks the loop by alternating: assume three of the four unknowns, update the fourth, and repeat.

01

STEP 1 · FFBS

Sample states X

Walk forward building an alpha table; walk backward drawing a fresh state sequence. Runs once per game.

02

STEP 2 · TRANSITIONS

Sample v

Count within-game transitions across all games into a K×K matrix. Each row ⟶ Dirichlet draw.

03

STEP 3 · INITIAL

Sample π

Count the first-pitch state across ~30 games. One Dirichlet draw gives a new opening distribution.

04

STEP 4 · EMISSION

Sample μ

Pool pitches by state across the season. Each μ□ ⟶ Normal draw, precision-weighted prior + data.

One full Gibbs iteration = one pass through all four blocks. Run 10,000 iterations × 4 chains, drop first 1,000 (burn-in), keep every 10th (thinning) ⟶ 3,600 pooled posterior draws.

Games are independent sequences; the state resets at every start.

APPLIED EXAMPLE

Step 1 · Sample the hidden states

STEP 1 · FFBS

Plain English

For each game, walk forward building an α table; “how plausible is each state given what I’ve seen so far?” α combines the previous α, the transition probability, and how well the velocity matches each state’s mean. Then walk backward sampling a full state sequence from α.

FULL CONDITIONAL

$$\alpha[1, i] = \pi_i \, N(y_t \mid \mu_i, \sigma)$$

$$\alpha[t, i] = \Sigma_j \; \alpha[t-1, j] \cdot v[j, i] \cdot N(y_t \mid \mu_i, \sigma)$$

$$X_t \sim \alpha[t, \cdot] \cdot v[\cdot, X_{t+1}] \quad \text{(backward)}$$

Runs once per game; each game has its own α table, initialized from π.

α TABLE · ONE GAME

Pitch t	Velocity (mph)	α[Baseline]	α[Hot]
1	95.8	0.0421	0.0018
2	96.2	0.0156	0.0009
3	97.4	0.0048	0.0012
4	98.9	0.0009	0.0011
5	98.6	0.0002	0.0007

CODE

```
alpha[t,i] <- alpha[t,i] + alpha[t-1,j] * curnu[j,i] * dnorm(Y[[g]][t],
curmu[i], sigma)
```

Values illustrative.

t — pitch within a game (1, 2, …, T_g)
g — game (1, …, G)
i — current state (the one we’re computing α for)
j — previous state (summed over)

APPLIED EXAMPLE

Step 2 · Count transitions, draw v

STEP 2 · TRANSITIONS v

Plain English

With X in hand, count within-game transitions across all games: Baseline→Baseline, Baseline→Hot. That fills a K×K count matrix N. Add the Dir(8, 1) prior to each row and draw a new row from a Dirichlet. Cross-game transitions are never counted.

FULL CONDITIONAL

$$N[j, k] = \#\{t : X_{t-1}=j,\ X_t=k\}$$

$$v[j, \cdot] \sim \text{Dirichlet}(N[j,1] + a[j,1], \ldots, N[j,K] + a[j,K])$$

A Dirichlet draw = Gamma draws, then normalize.

COUNTS N → NORMALIZED v

	→ Baseline	→ Hot	Row → v
From Baseline	812	94	(0.897, 0.103)
From Hot	97	443	(0.180, 0.820)

CODE

```
curnu[j,k] <- rgamma(1, shape = Nmat[j,k] + prior_nu[j,k], rate = 1)
```

Values illustrative.

t — pitch within a game
j — “from” state (row of N and v)
k — “to” state (column of N and v)

APPLIED EXAMPLE

Step 3 · Count first pitches, draw π

STEP 3 · INITIAL π

Plain English

Same idea as v, but only for the first pitch of each game. Across 17 starts, tally how many opened in Baseline vs. Hot, add the Dir(6, 2) prior, and draw a fresh π. Game independence is what gives π 17 data points instead of just 1.

FULL CONDITIONAL

$$n^{init}_j = \#\{g : X_1^{(g)} = j\}$$

$$\pi \sim \text{Dirichlet}(n^{init}_1 + b_1, \ldots, n^{init}_K + b_K)$$

INIT COUNTS → NORMALIZED π

State	Games starting here	π (after normalize)
Baseline	14	0.737
Hot	3	0.263

CODE

```
curpi[j] <- rgamma(1, shape = init_count[j] + prior_pi[j], rate = 1)
```

Values illustrative.

g — game (1, …, G = 17 starts)
j — state
K — total number of states (= 2 here)

APPLIED EXAMPLE

Step 4 · Pool pitches, draw a new mean

STEP 4 · EMISSION μ

Plain English

For each state k, pool every pitch assigned to it across all games and take the sample mean. The posterior mean of μ□ is a precision-weighted blend of the prior mean and the sample mean.

FULL CONDITIONAL

$$\text{post_prec} = 1/\tau^2 + n_k/\sigma^2$$

$$\text{post_mean} = (m_k/\tau^2 + n_k \cdot \bar{y}_k/\sigma^2) \;/\; \text{post_prec}$$

$$\mu_k \sim N(\text{post_mean},\ 1/\text{post_prec})$$

POOLED POSTERIOR · BY STATE

State	n_k pitches	Sample mean (mph)	Post. mean
Baseline	906	96.24	96.24
Hot	540	98.41	98.40

CODE

```
curmu[j] <- rnorm(1, mean = post_mean, sd = sqrt(1 / post_prec))
```

Values illustrative.

- **k** — state being updated
- **m_k** — prior mean for state k (96 or 99)
- **τ^2** — prior variance on μ_k
- **σ** — fixed emission SD (empirical SD of velocities)
- **n_k** — pitches assigned to state k (pooled across games)
- **$\bar{y}_k$** — sample mean velocity of those pooled pitches

APPLIED EXAMPLE

Label switching: one last fix

WHY IT HAPPENS

The labels are arbitrary

The likelihood is identical under a label swap; the model doesn't know which state we call "Baseline" and which we call "Hot." Across iterations, labels can flip at random.

WHY WE CARE

Averaging mixes the two states

Take the posterior mean of μ_1 across 3,600 draws and half the samples might be Hot values. You end up averaging Baseline and Hot together into garbage data.

THE FIX

After each iteration, enforce $\mu_{Baseline} < \mu_{Hot}$.

If the means are flipped, swap μ, swap π, swap both rows and columns of V, and relabel every X.

ONE ITERATION · BEFORE vs. AFTER

Quantity	Before fix (flipped)	After fix (sorted)
μ	(98.40, 96.24)	(96.24, 98.40)
π	(0.26, 0.74)	(0.74, 0.26)
V[1, ·]	(0.82, 0.18)	(0.90, 0.10)
V[2, ·]	(0.10, 0.90)	(0.18, 0.82)
X (one game)	[2, 2, 2, 1, 1, 1, …]	[1, 1, 1, 2, 2, 2, …]

HOW THE SWAP PROPAGATES

- μ: swap entries
- π: swap entries
- V: permute both rows AND columns — new V[i, j] = old V[σ(i), σ(j)]
- X: relabel every pitch in every game

Applied post-hoc to saved draws, not inside the sampler. Iterations where the order is already correct are left alone.

APPLIED EXAMPLE

MCMC diagnostics

*We ran **4 chains × 10,000** iterations from different starting values, then burned the first **1,000 and thinned by 10**. That leaves **900 draws per chain**, pooled across chains to **3,600 posterior samples**. Label switching is fixed post-hoc by sorting on μ at each iteration.*

01

EMISSION MEANS

μ_1, μ_2

Posterior distributions over the Baseline and Hot mean velocities (mph).

02

TRANSITION ROWS

$\nu_{11}, \nu_{12}, \nu_{21}, \nu_{22}$

All four cells of the transition matrix: stay-vs-switch probabilities from each state.

03

INITIAL STATE

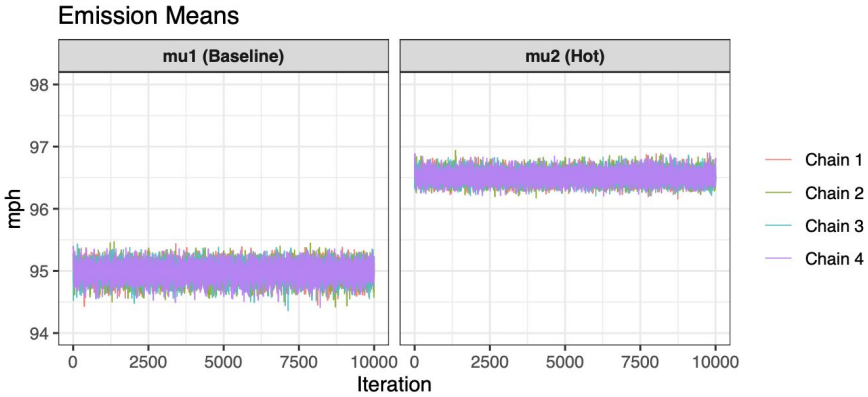
π_1, π_2

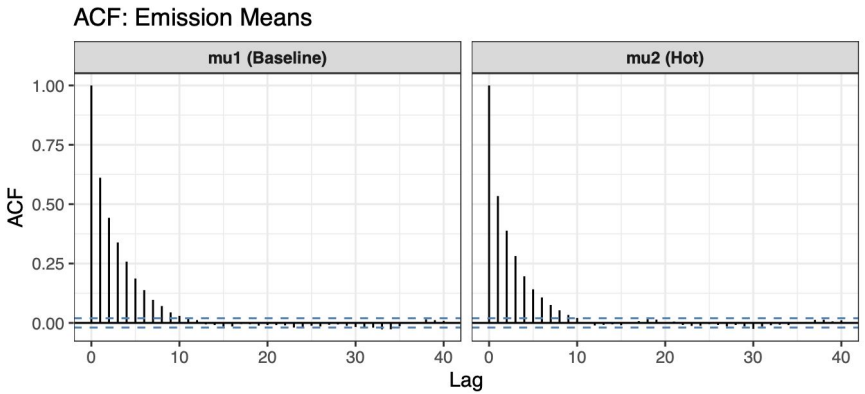
Posterior probability that a start begins in Baseline vs. Hot.

APPLIED EXAMPLE

Trace & ACF: Emission means

μ_1 (Baseline), μ_2 (Hot)

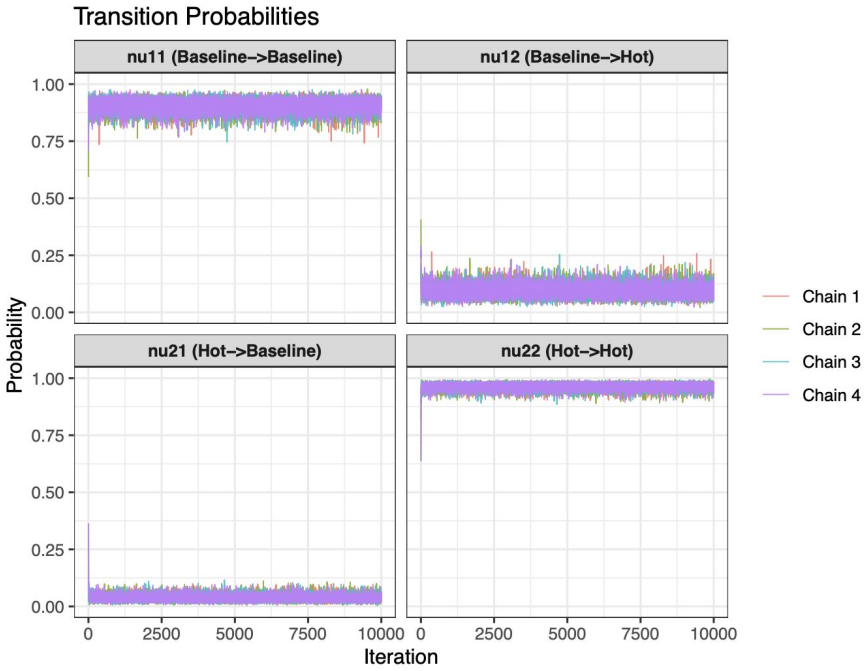


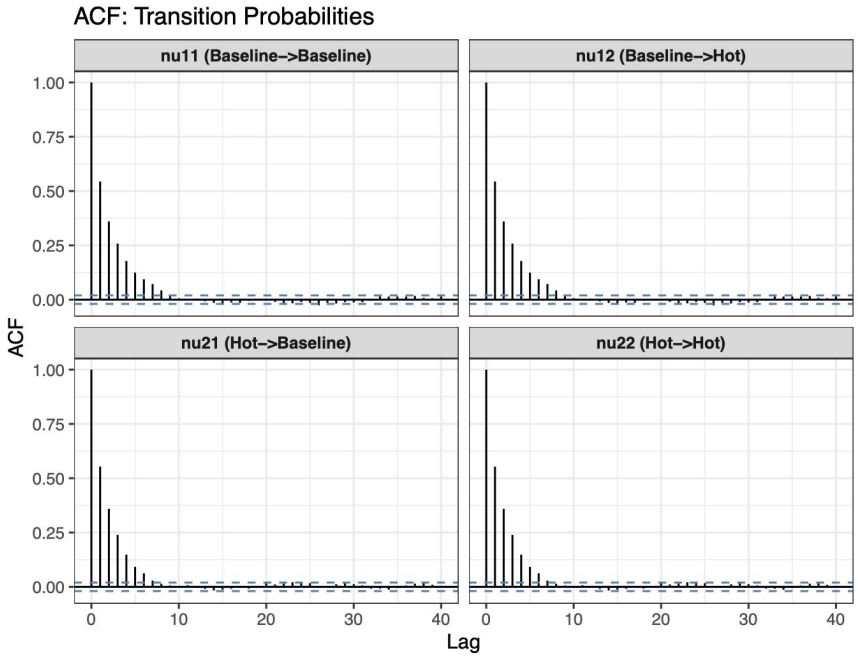


APPLIED EXAMPLE

Trace & ACF: Transition probabilities

v_{11}, v_{12}, v_{21}, v_{22} (four cells of the transition matrix)

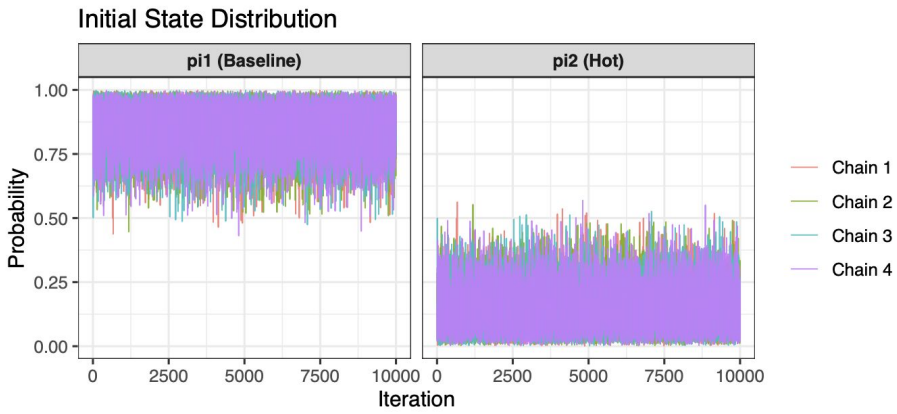


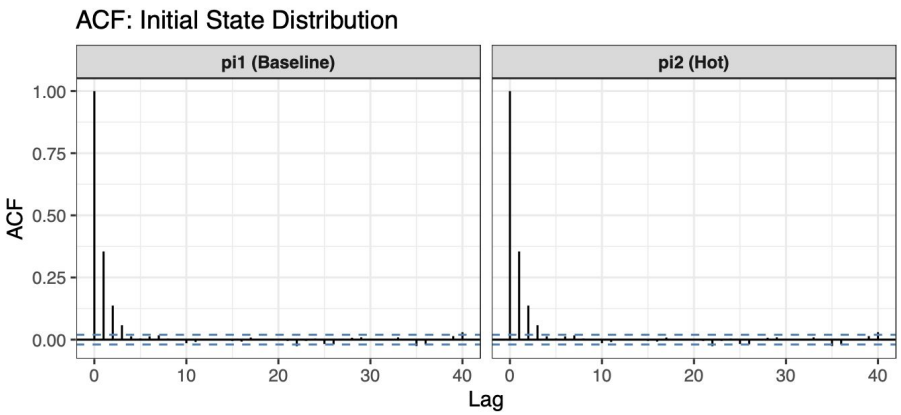


APPLIED EXAMPLE

Trace & ACF: Initial state

π_1 (Baseline start), π_2 (Hot start)





APPLIED EXAMPLE

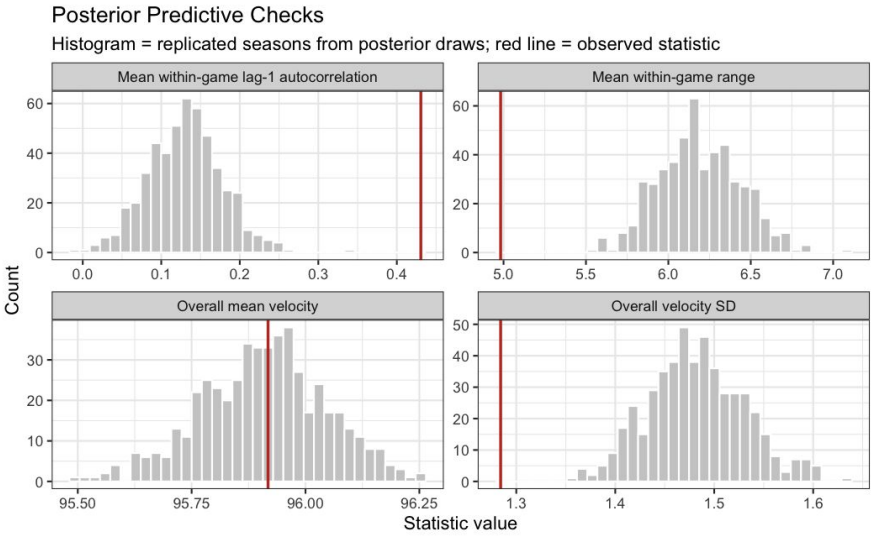
Posterior summaries

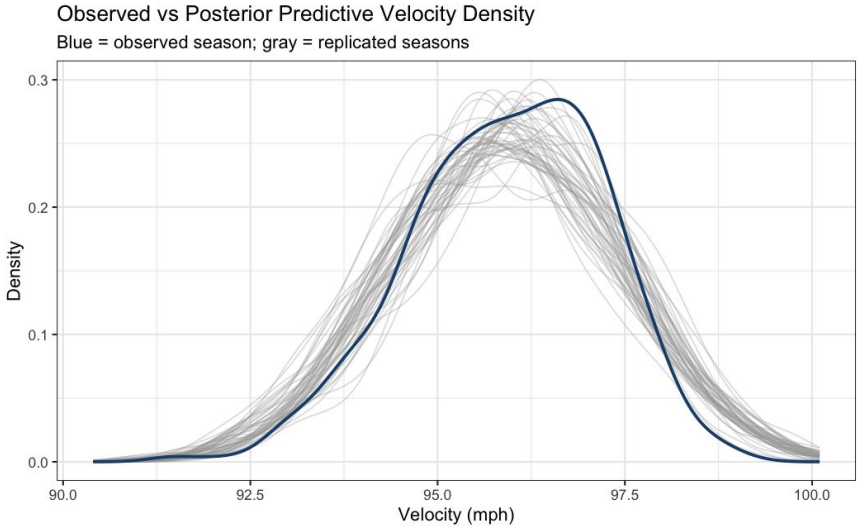
Pooled across 4 chains × 900 thinned draws = 3,600 samples. Intervals are 80% central credible intervals.

Parameter	Mean	SD	10%	90%
μ_1 Baseline mean (mph)	94.984	0.126	94.821	95.139
μ_2 Hot mean (mph)	96.517	0.092	96.401	96.638
v_{11} Baseline → Baseline	0.905	0.027	0.870	0.937
v_{12} Baseline → Hot	0.095	0.027	0.063	0.130
v_{21} Hot → Baseline	0.038	0.014	0.022	0.056
v_{22} Hot → Hot	0.962	0.014	0.944	0.978
π_1 Pr(start Baseline)	0.860	0.082	0.744	0.956
π_2 Pr(start Hot)	0.140	0.082	0.044	0.256

APPLIED EXAMPLE

Posterior Predictive Checks - Initial Attempt





Statistic	Observed	Replicated_Mean	Replicated_2.5	Replicated_97.5
Overall mean velocity	95.918	95.909	95.627	96.162
Overall velocity SD	1.284	1.482	1.394	1.587
Mean within-game range	4.982	6.189	5.725	6.677
Mean within-game lag-1 autocorrelation	0.431	0.131	0.040	0.222

APPLIED EXAMPLE

Potential Model Improvement

State-specific variances

- Allowed each latent state to have their own variability (σ_1 != σ_2)
- Fixes unrealistic assumption of constant noise across states

Within-game velocity trend

- Added linear effect of pitch number. Changed emission model:

$$y_t \sim N(u_{Zt} + \beta * t, \sigma_{Zt})$$

- Captures warm-up/fatigue dynamics within a start

Intuition

- Trend + HMM reduces need for excessive state switching
- Improves autocorrelation and within-game behavior
- More flexible data-generating process that separates state differences, within-state variation, and systematic trends

Code Changes

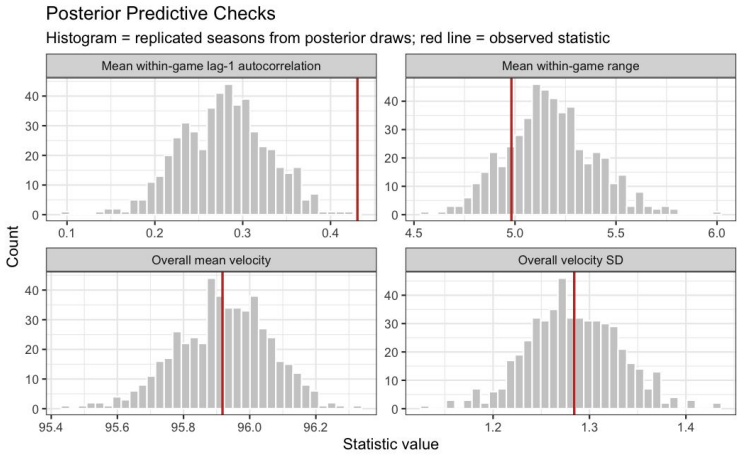
- New covariate input: Centered pitch-number vector for each game
- Gibbs sample: Added sampling step for linear trend and state-specific variance parameters
- Priors: Normal prior for beta and inverse gamma for variance terms
- Label switching: Swaps state-specific variances when labels get switched. Still reorders states by mean

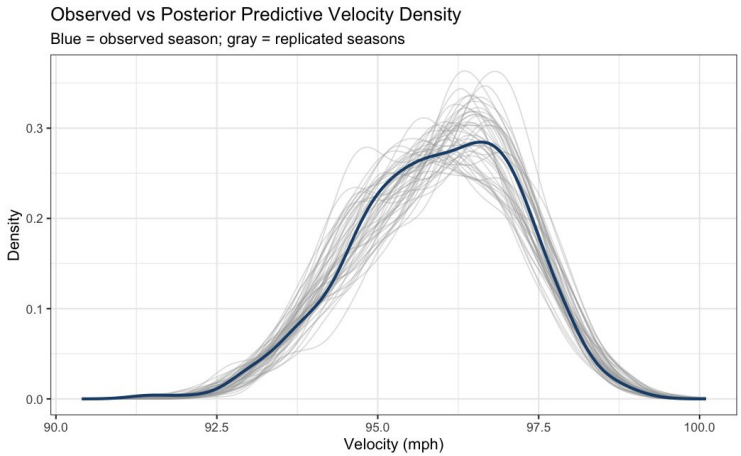
Interpretation would change here!

- Reflects performance regimes **after** accounting for systematic within-game trends and different noise levels

APPLIED EXAMPLE

Posterior Predictive Checks - After





Posterior summaries from the HMM with state-specific variances and a linear pitch-number trend.

Parameter	Mean	SD	10%	90%
mu1 (Baseline mean, mph)	95.014	0.117	94.859	95.161
mu2 (Hot mean, mph)	96.759	0.091	96.641	96.873
beta (pitch-number trend)	0.005	0.006	-0.004	0.013
sigma1^2 (Baseline variance)	1.077	0.102	0.947	1.207
sigma2^2 (Hot variance)	0.710	0.078	0.610	0.813
nu11 (Baseline -> Baseline)	0.877	0.029	0.839	0.912
nu12 (Baseline -> Hot)	0.123	0.029	0.088	0.161
nu21 (Hot -> Baseline)	0.091	0.022	0.063	0.121
nu22 (Hot -> Hot)	0.909	0.022	0.879	0.937
pi1 (Pr. start Baseline)	0.884	0.072	0.786	0.966
pi2 (Pr. start Hot)	0.116	0.072	0.034	0.214

Posterior predictive check for the HMM with state-specific variances and a linear pitch-number trend.

Statistic	Observed	Replicated_Mean	Replicated_2.5	Replicated_97.5
Overall mean velocity	95.918	95.920	95.629	96.176
Overall velocity SD	1.284	1.284	1.185	1.372
Mean within-game range	4.982	5.175	4.769	5.628
Mean within-game lag-1 autocorrelation	0.431	0.279	0.187	0.374

SECTION 04

Interpretation

Inferring latent performance states in MLB pitching

INTERPRETATION

Latent State Probabilities and Observed Data

Game 2 (Early Return from Injury)

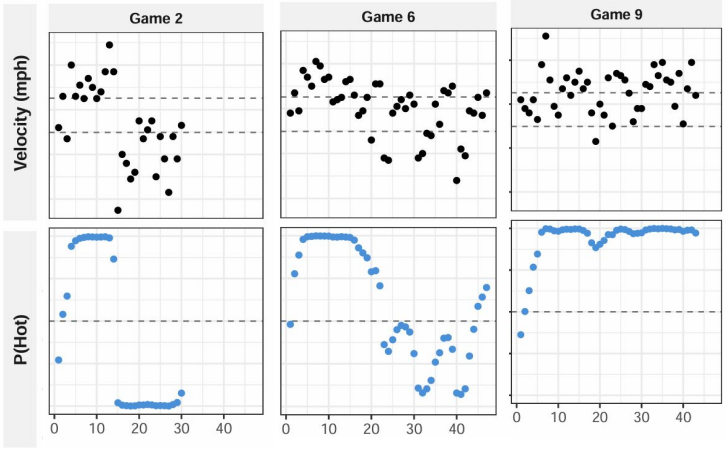
- Starts in a hot state, but drops sharply after ~15 pitches
- Indicates early fatigue / instability post-injury

Game 6 (Mid-Recovery)

- Begins hot, followed by a gradual decline
- Attempts recovery late, but never fully returns to hot state
- Suggests partial recovery but inconsistent stamina

Game 9 (Full Return to Form)

- Brief warm-up phase (first ~5 pitches)
- Sustains hot state for majority of game (~96 mph)
- Reflects stable, peak performance



Pitch number within game

MLB

Yankees' Gerrit Cole strikes out 10 in dominant outing while on 'pretty strict' pitch count

By Mark W. Sanchez

Published Aug. 11, 2024, 2:23 a.m. ET

39 Comments

There was plenty for the Yankees to be optimistic about after Gerrit Cole's most dominant outing of the season.

There was caution, though, because it was a shorter start than most.

Cole's stuff was excellent, with a season-high 10 strikeouts, on a night he only recorded 16 outs through 90 pitches.

The reigning AL Cy Young Award winner was on a "pretty strict" pitch count, manager Aaron Boone said, in being careful with an ace whose overall body soreness led to his spot being skipped in the rotation at the end of July.

Gerrit Cole walks off the mound in the sixth inning of the Yankees' second game against the Rangers on Aug. 10

JASON SZENES FOR THE NEW YORK POST

INTERPRETATION

Mean Posterior, 95% Confidence Interval

Average Hot State Over Game

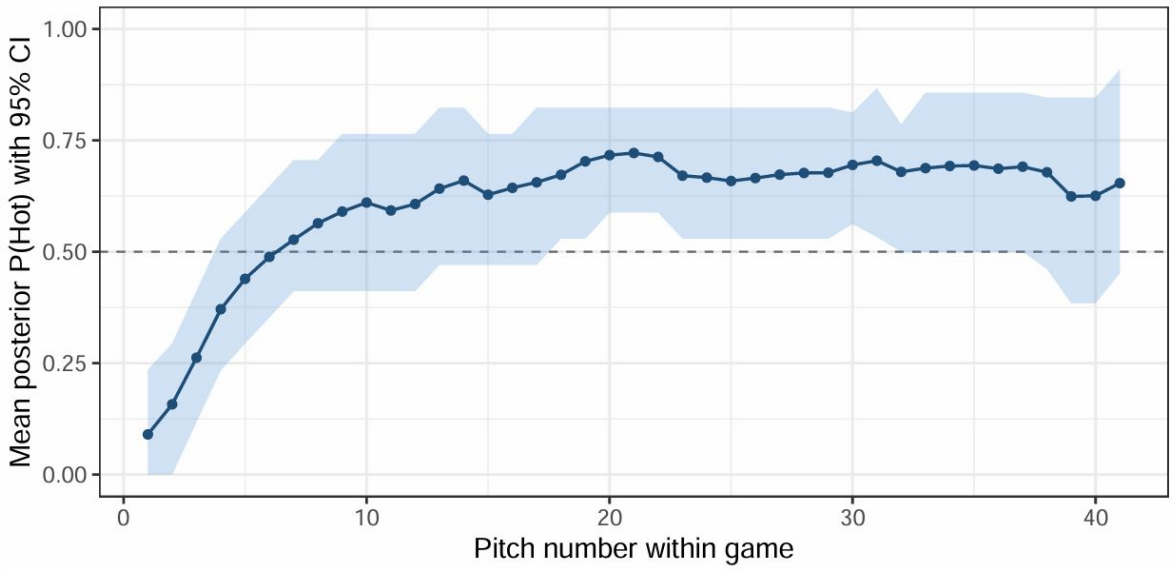
- X-axis: Pitch number within a game
- Y-axis: Mean posterior P(Hot) with 95% credible interval

How This Is Computed

- For each pitch position t, we average P(Hot) across all games
- Example:
  - Pitch 1 → average over all first pitches
  - Pitch 2 → average over all second pitches

Interpretation

- Cole typically starts in a baseline state early in games
- P(Hot) increases steadily as the game progresses
- Suggests a consistent warm-up pattern into peak performance



INTERPRETATION

What Did We Learn?

1 Core Findings

Our model identifies Baseline vs Hot regimes using velocity alone, with clear state persistence across pitches and meaningful mean velocity differences between states. However, the original model over-attributes variation to state-switching, producing unrealistic dynamics.

2 Improvements

Adding a within-game trend captures warm-up / fatigue. State-specific variances reflect consistency differences. The model now separates: state differences (mean velocity), within-state variability, and systematic game trends yielding substantially better fit and more realistic behavior.

3 Key Takeaway

Bayesian HMMs allow us to infer latent performance regimes from sequential data. Iterative model improvement informed by posterior diagnostics leads to richer, more credible inference about pitcher condition.

INTERPRETATION

Limitations: Our Model is an Oversimplification

1 Data & Observables

Single observable: only **velocity** is used, ignoring spin rate, movement, and other signals. We completely ignore external context: game conditions (inning, pitch count, opponent, weather) that may influence performance.

2 Computational Challenges

Gibbs sampling with multiple chains can be slow and resource-intensive. MCMC may mix slowly or get stuck, especially with latent states and tightly correlated parameters.

3 Modeling Sensitivity

Results can depend heavily on prior choices (e.g., how pre-2024 velocity splits are encoded). As previously mentioned, the two-state assumption imposes a binary interpretation. More nuanced regimes are compressed or lost.

INTERPRETATION

When Should We Use This Model?

1

Use When…

Data is sequential and driven by an unobserved evolving state. Goal is identifying latent regimes, not just average effects. Observations exhibit temporal dependence and persistence. Sequences can reasonably be treated as independent units (e.g., games).

2

Use the Original Model When…

You want a simple, interpretable model and the goal is to detect coarse regime changes without added complexity.

3

Use the Improved Model When…

There are within-sequence trends (warm-up, fatigue). Variability differs across regimes. Better fit and realism are more important than simplicity.

SECTION 05

Q&A

INTERPRETATION

Key pattern:

- If a pitcher starts in the baseline, he will likely stay in the baseline.
- In contrast, if a pitcher starts in the hot state, he will likely stay in the hot state.

Keybox representation:

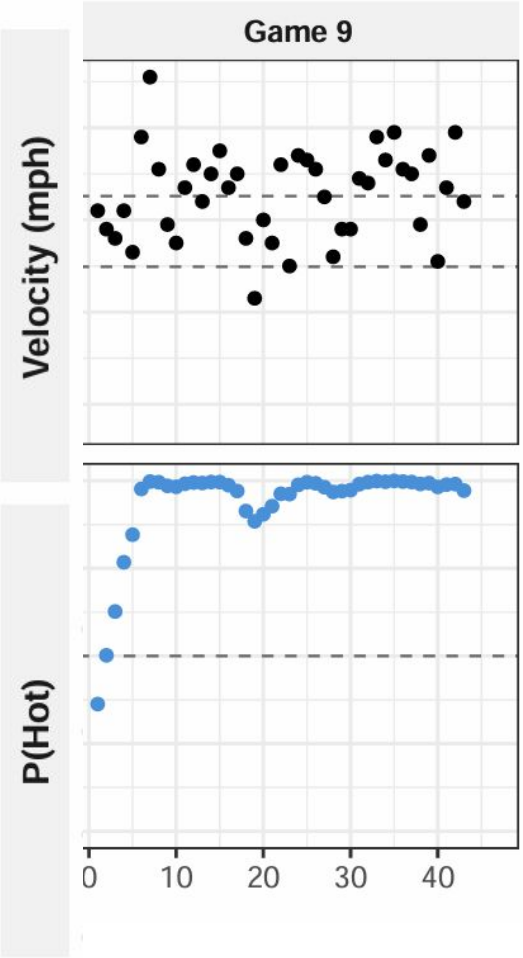
1. Black dots (velocity mph) are observed data.

   How fast Cole was throwing in sequence.

   VS

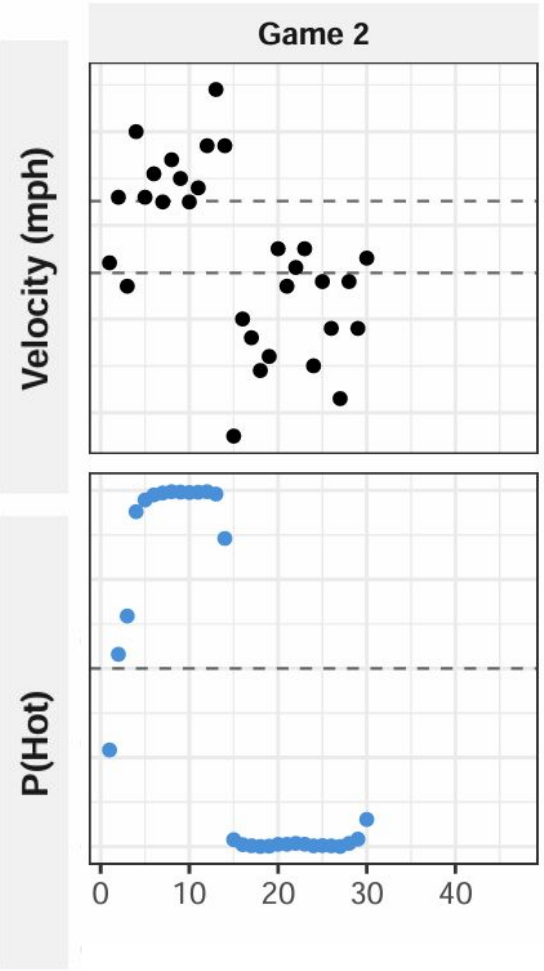
   P(Hot) state of Cole, which is our model that estimates Cole's hot state.

2. Definition: hotter state: a player is in a zone in the sport context and doing well in the field. If a player is in a hot state, that player will stay there on a conservative pitch by building momentum in terms of the number of pitches.



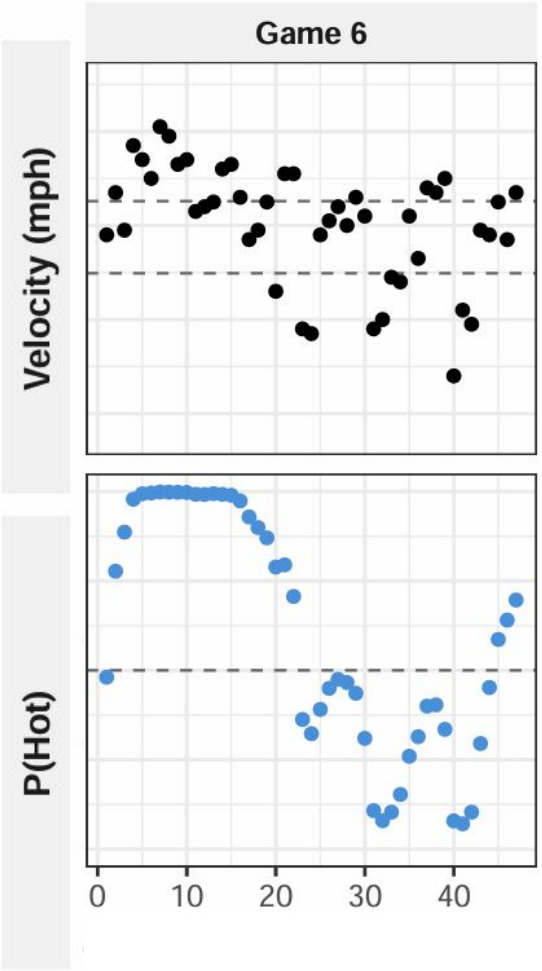
Game 2

- Game 2: This was Cole's second game after his injury. In the beginning, Cole was in a hot state, but l seemed to suddenly and significantly decrease his pitch after 15.



Game 6

- This was Cole's sixth game after his injury. In the beginning, he was in a hot state, but he decreased slowly after pitch 15. By pitch between 20 to 42, he did not achieve a hot state but he slowly tried to recover after pitch around 45 but never reached a full hot state as before in Game 6.

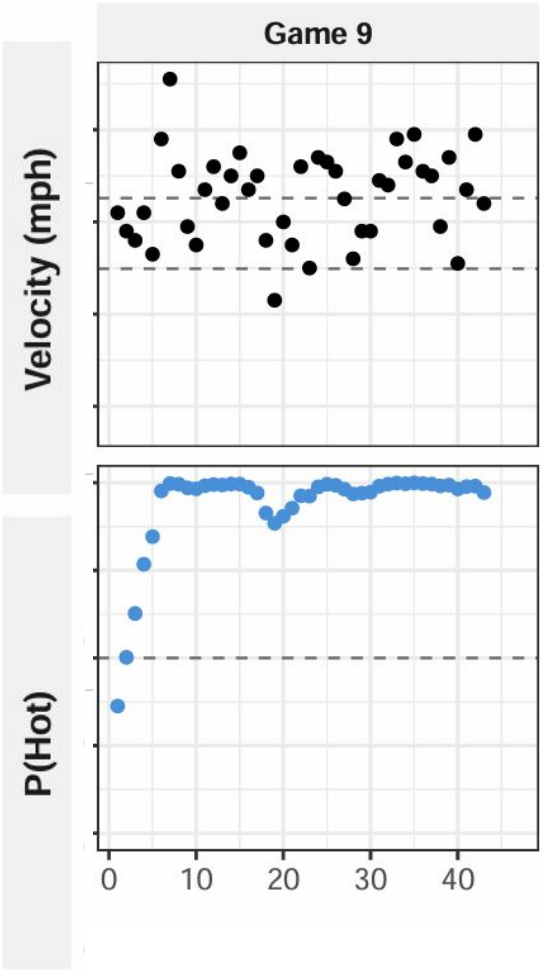


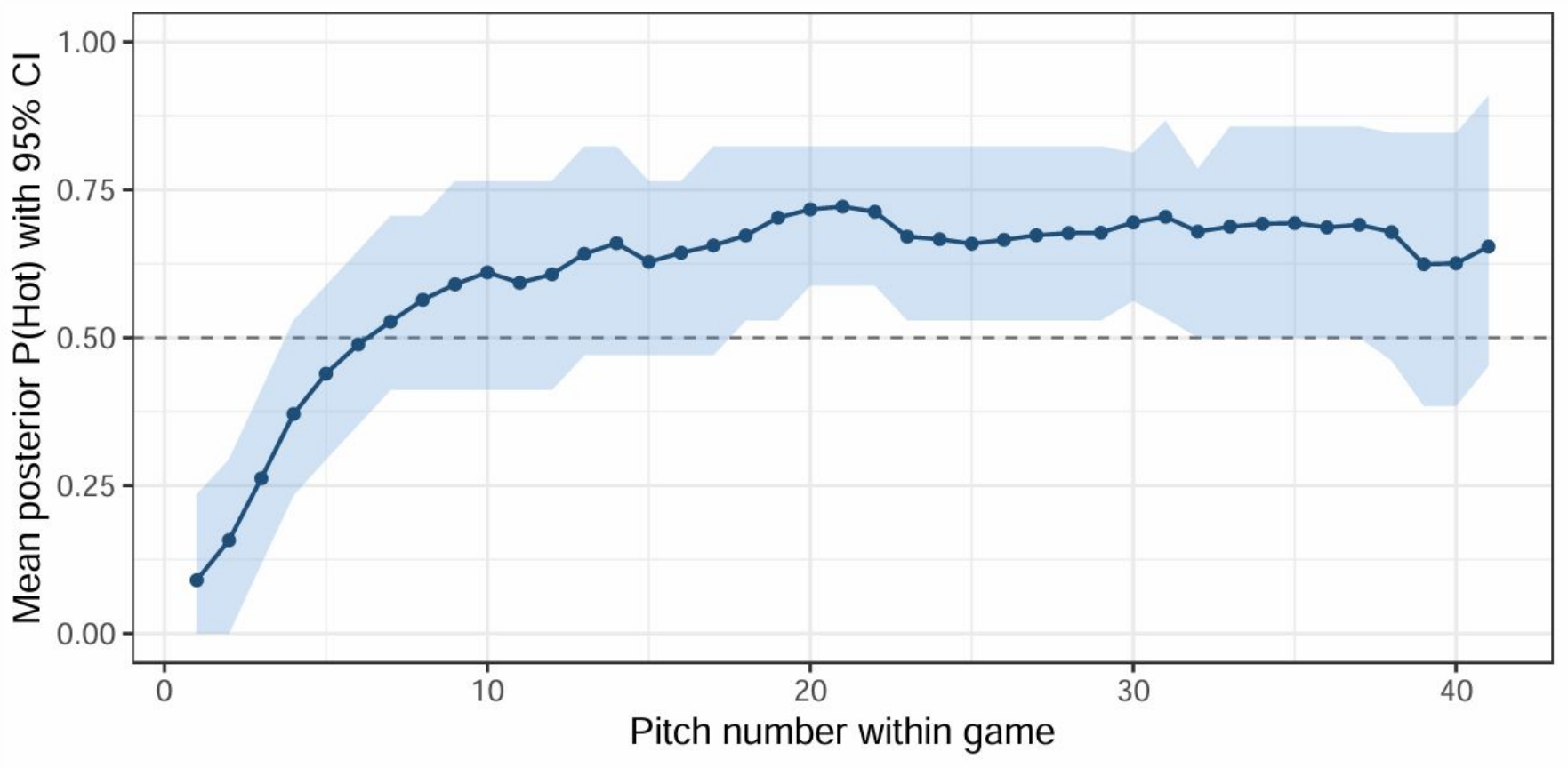
Game 9

- Game 9: This was Cole's 9th game after his injury, so he had more experience. By pitch between about 0 to 5, Cole warmed up. After pitch 5, he stayed in a hot state. Mr. Cole was pitching around 96 mph (miles per hour) for the whole game.

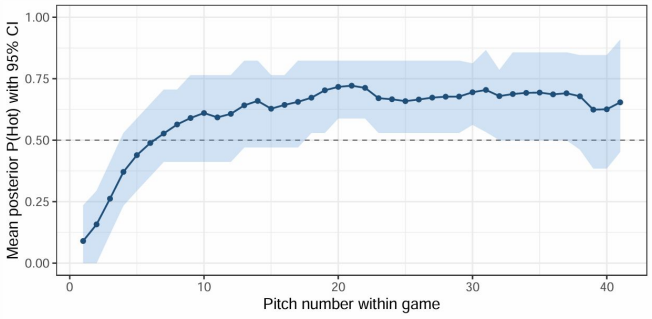
- A Key takeaway: Our model P(Hot) and velocity (mph) observed data are aligned in terms of results.

- Therefore, our model estimates that Mr. Cole was in a hot state. Our model aligns with the reality of Mr. Cole, who was in a hot state.




Mean posterior P(Hot) with 95% CI
1.00
0.75
0.50
0.25
0.00
0
10
20
30
40
Pitch number within game

our model averaged all pitches



- X-axis is “pitch number within game.”

  Y-axis is “mean posterior P(Hot) with 95% of confidence interval.”

- Over the course of the entire season, our model averaged all pitches that happened in the game.

- For Examples:

  1st pitch for all the game

  2nd pitch for all the game

We estimated based on the posterior that Mr. Cole started out as a baseline. As Mr. Cole proceeded with the game, he became hotter state.

Interpretation: “When should we use this model?”

1. Not cross-sectional data

2. Instead, regarding MCMC with the Gibbs sampler, use time series and panel data

Interpretation: “When should we use this model?”

1

If a pitcher starts in the baseline, he/she will likely stay in the baseline.

2

In contrast, if a pitcher starts in the hot state, he/she will likely stay in the hot state.

APPLIED EXAMPLE

Limitations

01

$\mu_{\square}$ ~ Normal

Center each state's mean velocity at a reasonable starting point, with enough spread to let the data dominate.

02

$v_{\square}$ ~ Dirichlet

Encode the belief that states are sticky. Most pitchers stay in the same state as the one before.

03

π ~ Dirichlet

Assume pitchers typically open a start in Baseline rather than already elevated.

Limitations: Our model is an oversimplification

1. Each data point had a sequence and a hidden state.

2. Sequence is dependent on hidden state (big assumption), so each game is independent.

3. Our model is an oversimplification of reality; We consider speed and pitch to determine whether a pitcher is in a zone or not.